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(1) Research Program/Project/Study Title: CAPSU Employee Time Tracking System with GPS	
Proponent Name:	JELLY L. PAREDES
Academic Rank:	ASSOCIATE PROFESSOR I
Co-Authors (with academic rank):	
Campus/Satellite College/Address:	MAMBUSAO SATELLITE COLLEGE
Telephone/Cell Phone/Fax Number:	09298111460
E-mail Address:	jlparedes@capsu.edu.ph

Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Mambusao
Province:	Capiz
Region:	VI

(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(7) Sector/Commodity	(8) Discipline/Field of Specialization Computer Science
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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

The Employee Time Tracking System with GPS is a comprehensive solution designed to accurately track and manage employee attendance, streamline leave management, and integrate GPS technology for location verification. This system utilizes advanced technology to automate time-tracking processes, improve accountability, and optimize resource allocation within CAPSU Mambusao Satellite College.

The implementation of the CAPSU Employee Time Tracking System with GPS is essential to ensure accurate and reliable tracking of employee attendance. Manual timekeeping methods can be prone to errors, leading to discrepancies and challenges in maintaining accountability. By utilizing GPS technology, the system provides precise location data, eliminating the possibility of fraudulent timekeeping practices and enhancing overall accountability.

The inclusion of GPS functionality in the time-tracking system promotes transparency among employees and supervisors. It allows for real-time monitoring of employee whereabouts during clock-ins and clock-outs, ensuring that employees are physically present at designated work locations. This fosters a sense of trust and transparency within the organization and discourages any unethical practices.

A comprehensive time tracking system enables CAPSU to effectively allocate resources and manage workloads. Accurate attendance data, combined with GPS location information, provides insights into employee productivity and time allocation. Supervisors can identify patterns, optimize resource distribution, and make informed decisions to ensure maximum efficiency and productivity within the institution.

The CAPSU Employee Time Tracking System with GPS generates comprehensive reports and analytics. These reports provide valuable insights into attendance patterns, and overall workforce management. The data can be analyzed to identify areas for improvement, optimize scheduling, and make informed decisions regarding resource allocation and productivity enhancement.

In today's evolving work environment, where remote and mobile work arrangements are increasingly common, the system's mobile accessibility is crucial. Employees can conveniently clock in and out, access relevant information through their mobile devices. This ensures that time tracking can be efficiently managed regardless of an employee's physical location.

By implementing the CAPSU Employee Time Tracking System with GPS, the institution can improve accuracy, accountability, transparency, and resource optimization. The system streamlines time and attendance management, provides valuable insights for decision-making, and accommodates the needs of a modern workforce. It aligns CAPSU Mambusao Satellite College with best practices in time management, ensuring efficiency, fairness, and improved productivity across the organization/institution.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned. .)

The CAPSU Employee Time Tracking System with GPS is a comprehensive software solution designed to streamline and automate the tracking of employee attendance and work hours within Capiz State University (CAPSU). This system utilizes GPS technology to accurately monitor employee movements and ensure transparency in recording their time and attendance.

Key Features:

GPS Tracking: The system incorporates GPS technology to track the real-time location of employees. This allows for accurate monitoring of their movements and ensures that they are present at the designated work locations.

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Time and Attendance Management: The system records and manages employee time and attendance data. It captures clock-in and clock-out times, breaks, and leaves, providing an accurate overview of each employee's work hours.

Geofencing: Geofencing capabilities enable the system to define virtual boundaries around specific work areas. It ensures that employees can only clock in or out when they are within the designated boundaries, preventing fraudulent time entries.

Mobile Accessibility: The system offers mobile accessibility, allowing employees to clock in and out using their smartphones or tablets. This ensures convenience and flexibility, especially for employees who work remotely or in the field.

Reporting and Analytics: The system generates comprehensive reports and analytics on employee attendance and work hours. This data can be used for payroll processing, productivity analysis, and identifying trends or patterns in employee attendance.

Notifications and Alerts: The system sends notifications and alerts to employees and administrators regarding clock-in reminders, breaks, and overtime hours. This helps in ensuring compliance with attendance policies and regulations.

Integration with Payroll System: The system seamlessly integrates with the existing payroll system, automating the calculation of salaries and ensuring accurate and timely payments based on the recorded work hours.

Data Security: The system ensures the security and confidentiality of employee data through robust data encryption and access control measures. It complies with relevant data protection regulations to safeguard sensitive information.

Benefits:

Streamlined Attendance Tracking: The system eliminates manual attendance tracking processes, reducing errors and saving administrative time.

Increased Transparency: The GPS tracking feature ensures accurate recording of employee work hours and prevents time theft or unauthorized absences.

Enhanced Productivity: By providing insights into employee attendance patterns and trends, the system helps in identifying areas for productivity improvement.

Cost Savings: The system optimizes payroll processing by automating calculations and reducing the risk of errors, resulting in cost savings for the organization.

Improved Compliance: The system helps in ensuring compliance with attendance policies and regulations, reducing compliance-related risks.

The CAPSU Employee Time Tracking System with GPS is a valuable tool for CAPSU to streamline attendance management, improve productivity, and enhance overall efficiency. By implementing this system, CAPSU can ensure accurate and transparent tracking of employee work hours, leading to better resource utilization and improved organizational performance.

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

The implementation of the CAPSU Employee Time Tracking System with GPS aims to address the following key problems:

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1. Time Tracking:

Manual time tracking methods are prone to errors, leading to discrepancies in recording and calculating employee work hours. This creates challenges in accurately compensating employees and may result in payroll disputes.

2. Lack of Accountability:

The absence of a systematic time-tracking system can contribute to a lack of accountability among employees. Without proper monitoring, there is a risk of time theft, unauthorized absences, and negatively impacting productivity and resource allocation.

3. Limited Location Verification:

The absence of GPS integration in the time tracking system hinders the ability to verify employee locations during clock-ins and clock-outs. This lack of location verification may result in inaccurate attendance records and difficulties in ensuring employees are present at designated work locations.

4. Lack of Real-time Data and Reporting:

The absence of a centralized system for time tracking and leave management limits the availability of real-time data and reporting. This hampers decision-making processes, as the supervisor/administrator may not have access to up-to-date information on attendance patterns, and resource allocation.

These problems hinder efficient time management, accountability, and resource optimization within CAPSU Mambusao Satellite College. Addressing these challenges through the implementation of the CAPSU Employee Time Tracking System with GPS will enable accurate time tracking, enhance accountability, ensure location verification, reduce administrative overhead, and provide real-time data for effective decision-making.

Objectives

Objectives of the CAPSU Employee Time Tracking System with GPS

1. create a system for CAPSU Employee Time Tracking System with the integration of GPS technology.
2. design an interface for accurately recording and tracking employee attendance, and physical presence at designated work locations during clock-ins and clock-outs.
3. develop a system that can generate real-time reports and provide a user-friendly interface for employees and administrator of CAPSU Mambusao Satellite College.
4. Evaluate a system using ISO 25010 standard (in terms of appearance, design, efficiency, and its operating performance).

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

In order to develop the CAPSU Employee Time Tracking System with GPS, a thorough review of existing literature and prior art related to time tracking systems, GPS integration, and employee management systems is crucial. This review helps identify existing technologies, approaches, and best practices that can inform the development and implementation of the system. The following sections highlight the key findings from the review of the literature and prior art search:

Time Tracking Systems

Several studies have explored the benefits of electronic time-tracking systems over traditional manual methods. Research indicates that electronic systems provide higher accuracy, reduce administrative workload, and improve overall efficiency in managing employee attendance (Chen et al., 2018). These systems offer features such as automated clock-ins and clock-outs, real-time monitoring, and automated data analysis (Kushwaha et al., 2019). The literature review also emphasizes the importance of integrating leave management modules into time-tracking systems to streamline the process and improve accuracy (Rahman et al., 2020).

GPS Integration

GPS technology has gained significant attention in various fields, including employee management systems.

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Studies have highlighted the benefits of GPS integration in time-tracking systems, particularly for remote and mobile workforce management (Chang et al., 2017). GPS-enabled systems provide real-time location data, ensuring employees' physical presence at designated work locations and reducing the risk of fraudulent practices (Lee et al., 2018). The literature also emphasizes the importance of privacy considerations and adherence to relevant regulations when implementing GPS tracking systems (Ma et al., 2020).

Employee Management Systems

Literature on employee management systems highlights the importance of comprehensive solutions that go beyond time tracking. Integrated systems that include features such as leave management, performance evaluation, and resource allocation have been shown to improve overall productivity and employee satisfaction (Kumar et al., 2019). The literature also emphasizes the need for user-friendly interfaces, mobile accessibility, and robust reporting and analytics capabilities (Nakamoto et al., 2020).

Prior Art Search

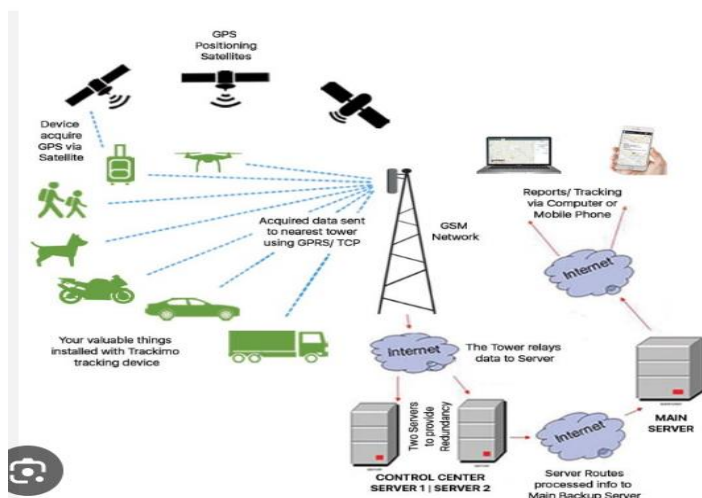
A comprehensive prior art search reveals existing technologies and systems related to employee time tracking and GPS integration. Several commercial solutions are available in the market, offering features such as mobile accessibility, GPS-based clock-ins, and real-time monitoring. Analyzing and benchmarking these prior art solutions can provide insights into their strengths, weaknesses, and potential areas for improvement.

Overall, the review of literature and prior art search indicates the significance of electronic time tracking systems, GPS integration, and comprehensive employee management solutions. The CAPSU Employee Time Tracking System with GPS can leverage these findings to design a customized system that addresses the specific requirements and challenges faced by CAPSU. By incorporating best practices and innovative features, the system aims to enhance accuracy, accountability, and productivity within the institution.

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

Theoretical Framework for CAPSU Employee Time Tracking System with GPS.

The theoretical framework provides a foundation for understanding the concepts, principles, and theories that underpin the CAPSU Employee Time Tracking System with GPS. It helps guide the development, implementation, and evaluation of the system. The following theoretical perspectives can be considered for the project:



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Time Management Theories:

Time management theories offer insights into how individuals allocate, prioritize, and utilize their time effectively. The application of these theories can inform the design and features of the time tracking system. The literature on time management emphasizes concepts such as goal setting, task prioritization, and self-regulation (e.g., goal setting theory, time management matrix). These theories highlight the importance of providing tools and features in the system that facilitate effective time management practices for employees.

Technology Acceptance Model (TAM):

The Technology Acceptance Model (TAM) can be employed to understand employees' attitudes and perceptions towards the new time tracking system. TAM examines the factors that influence individuals' acceptance and usage of technology. It considers elements such as perceived usefulness, ease of use, and individual attitudes towards technology adoption. By incorporating the principles of TAM, the system can be designed to maximize user acceptance and engagement among CAPSU employees.

Employee Engagement Theory:

Employee engagement theory focuses on the psychological and emotional commitment of employees towards their work and organization. By incorporating features that enhance employee engagement, the time tracking system can foster a sense of ownership, motivation, and involvement among employees. The literature on employee engagement emphasizes factors such as autonomy, clear expectations, and recognition. The system can provide features that enable employees to have autonomy in managing their time, receive timely feedback, and be recognized for their efforts.

Location-Based Services (LBS) Theories:

As the time tracking system incorporates GPS technology, theories related to Location-Based Services (LBS) can be considered. LBS theories examine the impact of location information on user behavior, decision-making, and privacy concerns. By considering LBS theories, the system can ensure that GPS functionality is implemented in a manner that respects privacy, provides value to employees and supervisors, and addresses any potential ethical or legal concerns.

Organizational Change Theories:

Implementing a new system such as the CAPSU Employee Time Tracking System with GPS requires managing organizational change. The literature on organizational change theories can provide insights into managing the transition effectively. Theories such as the Kotter's 8-Step Change Model or Lewin's Change Management Model can guide the implementation process, including communication, training, and addressing resistance to change.

By drawing from these theoretical perspectives, the CAPSU Employee Time Tracking System with GPS can be developed and implemented with a strong theoretical foundation. The integration of time management theories, technology acceptance models, employee engagement theories, location-based services theories, and organizational change theories will ensure that the system meets the needs of CAPSU employees, enhances productivity, and drives positive organizational outcomes.

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

The Software Development Life Cycle (SDLC) for the methodology of implementing the CAPSU Employee Time Tracking System with GPS can be structured as follows:

1. Planning (Requirements Gathering)

Identify and document the specific requirements and objectives of the system based on CAPSU's needs assessment.

Gather input from stakeholders, including employees, supervisors, and HR personnel, to ensure all relevant requirements are captured.

2. Designing (System Design)

Design the system architecture, user interface, and database structure based on the identified

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requirements.
 Define the functionalities and features of the system, including clock-in/out, leave management, GPS integration, reporting, and user roles.
 Create a detailed design document outlining the system components, modules, and their interactions.

3. Coding (System Development)
 Develop the Employee Time Tracking System with GPS based on the design specifications.
 Follow coding best practices, maintain coding standards, and ensure the system is modular, scalable, and maintainable.
 Conduct regular code reviews and testing to identify and fix any bugs or issues during the development process.

4. Testing (System Testing)
 Perform different types of testing, such as unit testing, integration testing, system testing, and user acceptance testing.
 Validate that the system meets the defined requirements and functions as expected.
 Identify and resolve any defects or issues discovered during testing.

5. Deployment (Releasing)
 Prepare the system for deployment by configuring it according to CAPSU's infrastructure and requirements.
 Plan the deployment strategy, including any necessary hardware or software installations.
 Conduct a pilot deployment in a controlled environment to ensure system stability and compatibility.

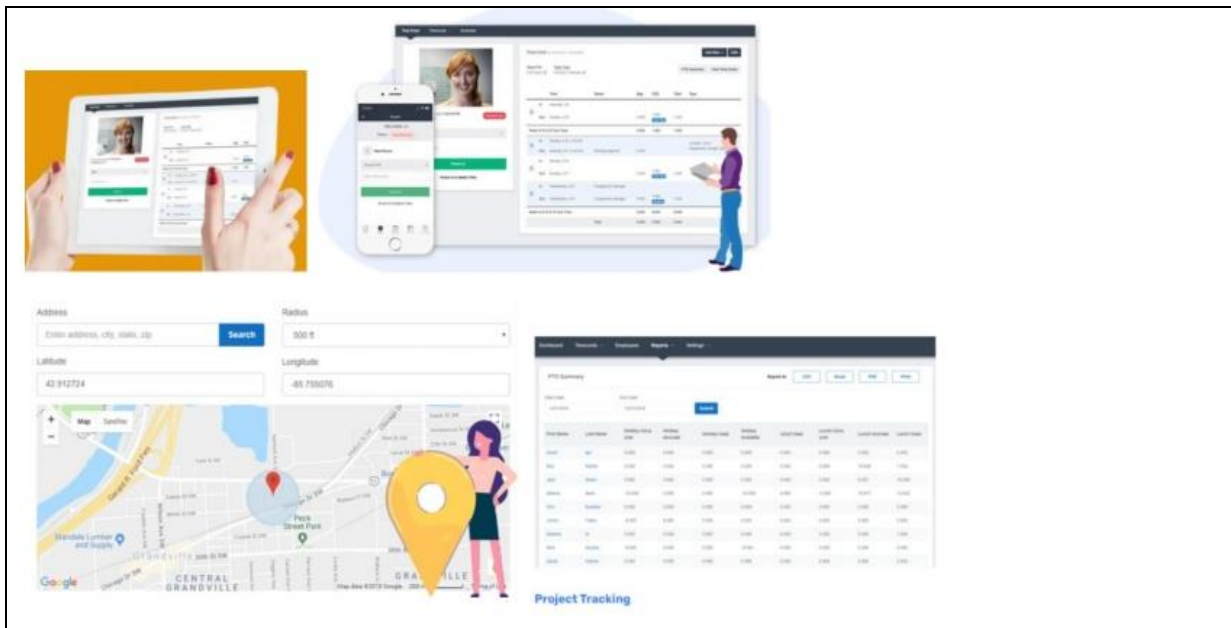
By following this SDLC methodology, the implementation of the CAPSU Employee Time Tracking System with GPS will be carried out systematically, ensuring a structured approach to development, testing, deployment, and ongoing maintenance.

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

Research Objectives	Year 1						Year 2						
	July	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	
1. Planning													
2. Designing													
3. Coding													
4. Testing													

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

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(16) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

(17) Target Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

The CAPSU Employee Time Tracking System with GPS is designed to benefit various stakeholders within CAPSU, including:

Employees:

Employees across all departments and levels within CAPSU will be beneficiaries of the system. They will have access to a user-friendly interface to clock in/out, request leaves, and view their attendance records. The system will provide employees with accurate and transparent time tracking, ensuring fair compensation for their work and promoting accountability.

Supervisors and Administrators:

Supervisors and administrators will benefit from the system's real-time monitoring dashboard, which provides a centralized view of employee attendance, GPS locations, and leave requests. This feature enables supervisors to effectively manage their teams, track employee productivity, and make informed decisions regarding resource allocation and scheduling.

CAPSU Management:

The management team at CAPSU will benefit from the system's comprehensive reports and analytics. These reports provide valuable insights into attendance trends, leave utilization, and productivity, assisting in strategic decision-making and resource planning. The system enables management to optimize workforce management and improve overall operational efficiency.

IT Department:

The IT department within CAPSU will play a critical role in the implementation and maintenance of the system. They will benefit from the system's utilization of modern technologies, including GPS integration and mobile accessibility. The system will provide opportunities for IT professionals to enhance their technical skills and contribute to the digital transformation initiatives at CAPSU.

CAPSU as an Institution:

The implementation of the CAPSU Employee Time Tracking System with GPS will have a positive impact on CAPSU as an institution. The system promotes a culture of efficiency, accountability, and transparency. It helps streamline administrative processes, reduce manual errors, and enhance overall productivity. By optimizing

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workforce management, CAPSU can achieve cost savings, improve resource allocation, and ensure fair compensation for employees.

In summary, the target beneficiaries of the CAPSU Employee Time Tracking System with GPS include employees, supervisors, administrators, the HR department, CAPSU management, the IT department, and CAPSU as an institution. The system aims to improve time management, accountability, and productivity, benefiting various stakeholders involved in the daily operations of CAPSU.

(18) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(19) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A Hardware Development Cost		
Web Server System Unit	1	P70,000.00
Wireless Keyboard and Mouse	1	P1,5000.00
Printer(EPSON L3110)	2	P8,000.00
UPS	2	P10,500.00
B. Server Development Cost		
Programmer	1	P120,000.00
GUI Designer	1	P80,000.00
Researcher	2	P60,000.00
Total		P260,000.00

(20) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

References:

Ajzen, I. (1991). The theory of planned behavior. Organizational Behavior and Human Decision Processes, 50(2), 179-211.

Chang, C., & Hsu, Y. (2017). Development of an employee location-based service system with GPS function. In Proceedings of the 2017 International Conference on Applied System Innovation (ICASI) (pp. 456-459).

Chen, W., Li, T., Huang, S., & Lin, L. (2018). An automatic employee attendance tracking system based on facial recognition and location services. Sustainability, 10(4), 1102.

Davenport, T. H., & Beck, J. C. (2001). The attention economy: Understanding the new currency of business. Harvard Business Press.

Kotter, J. P. (1996). Leading change. Harvard Business Review Press.

Kumar, A., Prakash, M., & Shukla, A. (2019). An employee management system using face recognition and fingerprint recognition techniques. International Journal of Engineering Research & Technology, 8(4),

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1384-1388.

Kushwaha, A. K., Kumar, S., Kumar, A., & Kumar, A. (2019). Design and implementation of employee attendance management system using biometric device.

Lewin, K. (1951). Field theory in social science: Selected theoretical papers. Harper & Row.

Venkatesh, V., & Davis, F. D. (2000). A theoretical extension of the technology acceptance model: Four longitudinal field studies. Management Science, 46(2), 186-204.

(21) Capsule Curriculum Vitae (One-page brief C.V. only)

  +09198111460  jellyparedes36@gmail.com  Bgy. Tugus, Mambusao, Capiz  Jelly L. Paredes PERSONAL INFORMATION Date of Birth: September 1, 1980 Age: 43 Civil Status: Married Nationality: Filipino	JELLY L. PAREDES Associate Professor I ACADEMIC BACKGROUND 2017 – PRESENT Doctor of Education Major in Educational Management 54 units Capiz State University Burias, Campus Barangay Burias, Mambusao, Capiz 2010 – 2014 Master of Science in Computer Science ISAT-U Lapaz, Iloilo City 1998 – 2001 Bachelor of Science in Computer Science Colegio de la Purisima Concepcion Roxas City WORK EXPERIENCE 2004 – PRESENT FACULTY CAPIZ STATE UNIVERSITY MAMBUSAO SATELLITE COLLEGE TRAINING/SEMINARS: • National Certificate in Computer System Servicing • The Art of Writing and Publishing ICT Research Paper • Training-Workshop on Public Speaking • International Web Training on WEBEX • Philippine Society of IT Educators (PSITE) 24th National Convention • Data Privacy & Technology Start-ups Venture creation and Business model generation • Training on Intellectual property basics for CAPSU Faculty researchers and students. • Seminar-Workshop on Computer Interaction • SUC/TEA VI Webinar Series I • Webinar on Flexible Learning and Dynamic Learning Program (By: CHED R6) • Google Classroom: Refresher Course for Educators • Face to Facebook Classroom Instruction: Redefining social media as Remote Teaching and Learning Tools • Distance and Correspondence Education for Low Bandwidth and Low-Tech Remote Areas • Fundamentals of Design Thinking Process • Challenges and Issues in Flexible and Distance Learning • Innovations in Telecommunications • Legal Issues in Flexible and Distance Learning on Copyright • Quality Assurance in Blended and Distance Learning
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